

Key Terms in this Chapter

Term	Description
Associative dimensions	A mode of dimensioning that automatically updates dimension values when the dimension size is changed.
Benchmark	A reference point from where all distances are measured.
Spline	A flexible curve that can be drawn to fit any shape.
Hatch patterns	Ready-made patterns that can be added to a specified area in a drawing.
Justification	The alignment of text strings in a specific direction such as to the right, left, or center.
Polyline	A string of lines that may contain a number of line segments connected together.
Tangent	A point on a circle that forms 90° angle between the center point of the circle and another point outside the circle.

The Drawing Tools of CADD

The following are the basic drawing tools found in a CADD program:

- Line types
- Multiple parallel lines
- Flexible curves
- Arcs and circles
- Ellipses and elliptical arcs
- Text
- Dimensions
- Hatch patterns
- Polygons
- Arrows

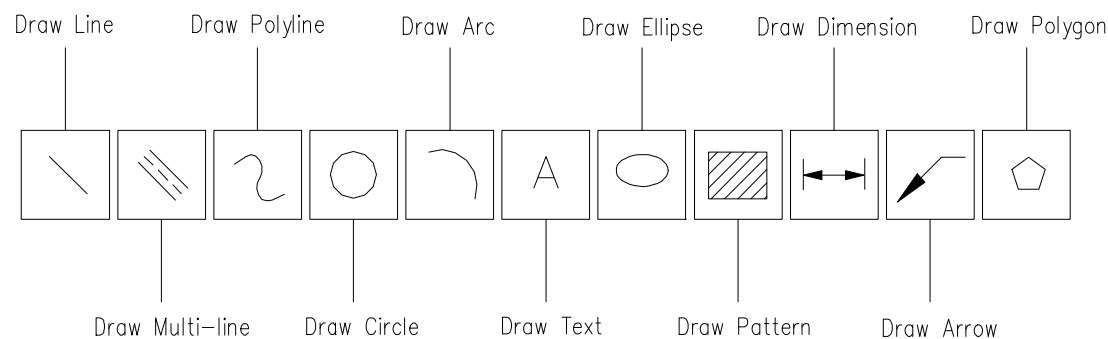


Fig. 3.1: Some examples of drawing tool buttons used in CADD.

Using Line Types

There are a number of line types available in CADD that can be used to enhance drawings. There are solid lines (or continuous lines), dash lines, dotted lines, center lines, etc. Fig. 3.2 shows some examples of common line types used in architectural and engineering drawings.

CADD enables you to follow both geometrical and engineering drawing standards. You can use line types to represent different elements in a drawing. For example, in engineering drawings, line types are commonly used to show different building services. In a plumbing plan, one line type can be used to show cold water pipes, one line type for hot water pipes, one line type for waste lines, etc.

CADD is preset to draw solid lines. When you enter the line command and indicate a starting point and end point, a solid line is drawn. If you want to draw with another line type, you need to set that line type as the default. Thereafter, all the lines are drawn with the newly selected line type.

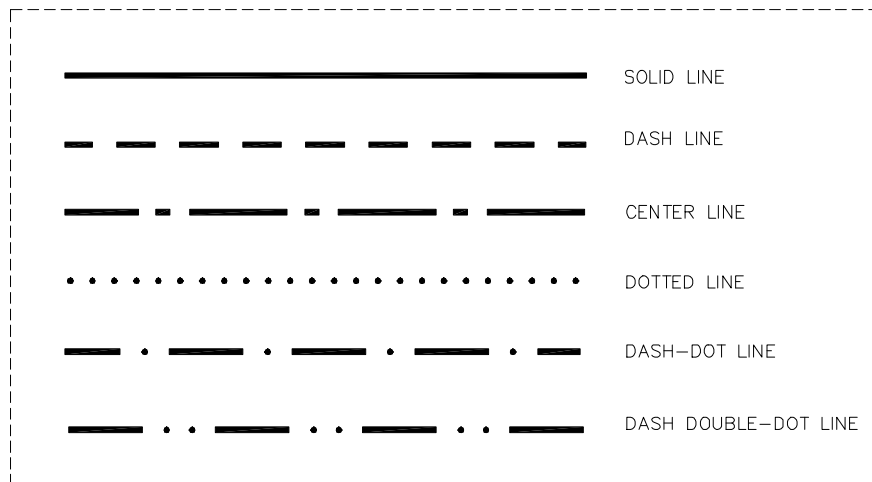


Fig. 3.2: Common line types used in architectural and engineering drawings.

Notes:

- International Standards Organization (ISO) has established standard names and definitions for line types. These standards are commonly followed by architects and engineers. US National CAD standard has also established independent standards for line types. For more information visit their web site at <http://www.nationalcadstandard.org>.
- You can change the line type of an existing line using CADD's edit functions. Some people prefer to first draw everything with one line type and later change line types as needed; others prefer to draw everything with the right line type the first time.

Drawing Multiple Parallel Lines

CADD allows you draw parallel lines (multi-lines) simultaneously just by indicating a starting point and an end point. For example, they can be used to draw the walls of a building plan, roads of a site map, or for any other presentation that requires parallel lines. Fig. 3.3 shows an example of drawing roads using multi-lines.

Most programs allow you to define a style for multiple parallel lines. You can specify how many parallel lines you need, at what distance and if they are to be filled with a pattern or shade.

Specialized CADD programs allow you to add attributes to multi-lines. For example, many architectural CADD software include a special function called "wall". When you use this function, it automatically draws parallel lines representing walls of specified style and thickness. When you draw a wall, it automatically clears all corners and intersections formed by walls. Advanced CADD programs even allow you to add attributes to the walls such as what kind of wall it is, material, price per linear foot, etc. The computer can keep track of all the walls in a drawing and can prepare a cost estimate using the database. Similar programs allow you to keep track of roadways, pipes, wires, etc. in different engineering disciplines.

Note:

Multiple parallel lines are a unified entity. Even though multiple lines are drawn, they are treated as one line. You cannot erase or edit one line separately. However, there are functions available that can break the entities apart.

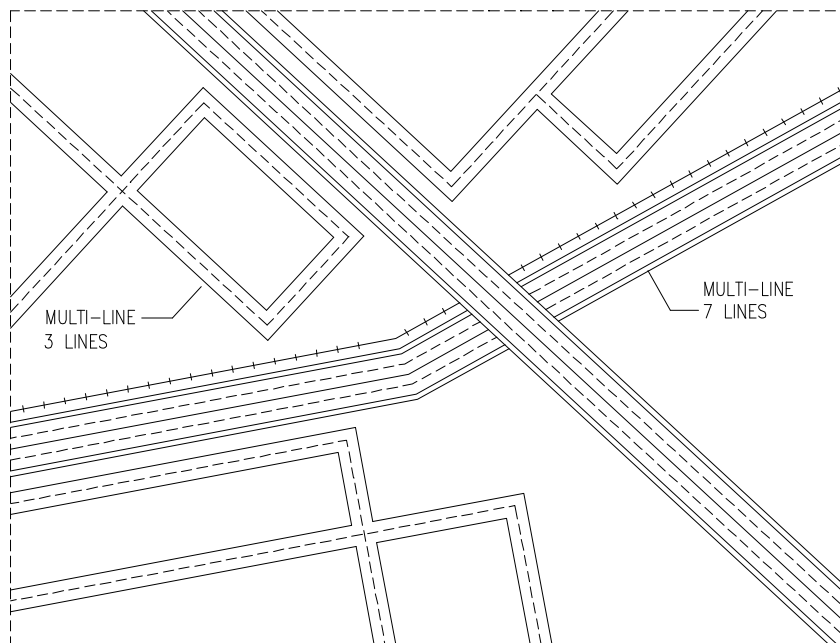


Fig. 3.3: Drawing a road layout using multiple parallel lines.

Drawing Flexible Curves

CADD allows you to draw flexible curves (often called splines) that can be used to draw almost any shape. They can be used to create the smooth curves of a sculpture, contours of a landscape plan or roads and boundaries of a map.

To draw a flexible curve, you need to indicate the points through which the curve will pass. A uniform curve is drawn passing through the indicated points. The sharpness of the curves, the roughness of the lines and the thickness can be controlled through the use of related commands.

Diagram A (Fig. 3.4) shows how you can draw a flexible curve by indicating points "1" through "9". You can edit the sharpness of curves and the thickness using the edit functions.

Diagram B (Fig. 3.4) shows how you can draw the contours of a landscape plan using flexible curves. You need to draw seven curves individually to complete the diagram. One of the contours in the diagram shows the actual points used to draw the curve.

Note:

The computer needs a lot of memory to draw flexible curves; they should be used only when necessary.

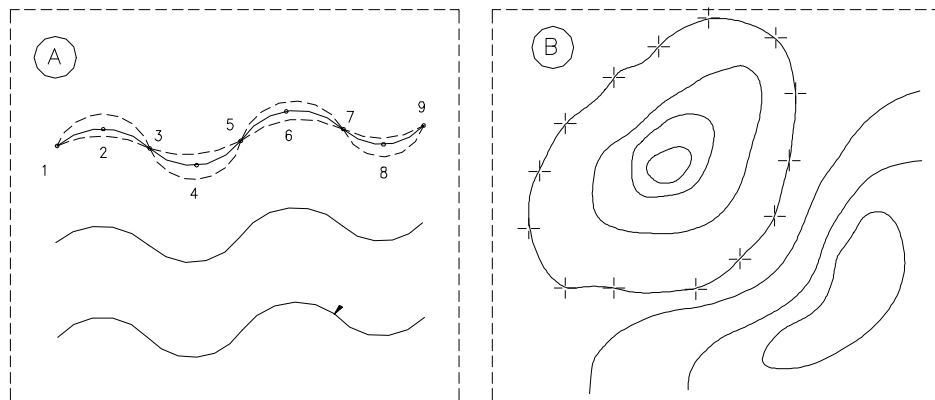


Fig. 3.4: *Drawing flexible curves.*

Important Tip:

There are add-on programs available for surveyors and landscape designers that facilitate drawing contours. These programs also allow you to add intelligent data to the contours such as dimensions from a benchmark and elevations from a datum line. This data can be stored in a database to prepare reports.

Drawing Arcs and Circles

CADD provides many ways to draw arcs and circles. There are a number of advanced techniques available for drawing arcs and circles, which can simplify many geometrical drawing problems. You can draw an arc by specifying circumference and radius, radius and rotation angle, chord length and radius, etc.

Arcs are drawn so accurately that a number of engineering problems can be solved graphically rather than mathematically. Suppose you need to measure the circumference of an arc, just select that arc and the exact value is displayed.

*The following are basic methods for drawing arcs and circles:
(These are essentially the same methods you learn in a geometry class.
However, when drawing with CADD the approach is a little different.)*

- Center point and radius
- 3 points
- Angle and radius
- 2 points
- 2 tangents and a point
- 3 tangents

Center Point and Radius

You can draw an arc or circle by indicating a center point and radius value. The center point can be entered by locating a point in the drawing area. The radius value can be entered numerically using the keyboard or by indicating a point in the drawing area. For example, to enter 1/2" as the radius, you can enter 1/2" using the keyboard or you can indicate a point in the drawing area that is 1/2" away from the selected center point.

To draw an arc, you need to specify two additional parameters: The starting point and the end point of the arc. When you locate these points in the drawing area, an arc is drawn from the starting point to the end point. Diagram A (Fig. 3.5) shows some examples of drawing arcs using this method. The center point of the arc is entered at "1", the starting point at "2" and the end point at "3".

Note:

The arcs are drawn in a counter-clockwise direction from starting point to end point. This is because CADD systems are generally set up to measure the counter-clockwise direction as positive. However, some programs provide the option to set the clockwise direction as the positive direction as well. In doing so, the arcs are drawn the opposite way.

3 Points

You can draw an arc or circle that passes through three given points. You need to enter three points through which the arc or circle will pass. Diagram B (Fig. 3.5) shows some examples of drawing arcs and a circle using this method. The points "1", "2" and "3" are used to draw the arcs and the circle.

Note:

If a circle or arc can't be drawn through the three points you entered, the computer will display an error message to that effect.

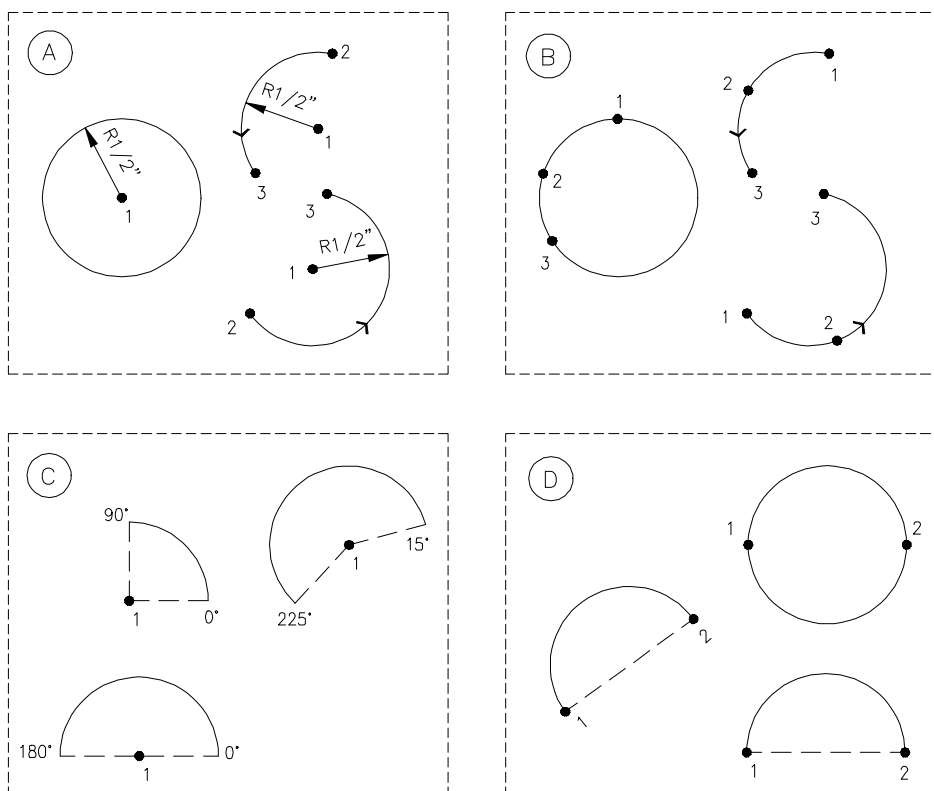


Fig. 3.5: Drawing arcs and circles.

Radius and Rotation Angle

You can draw an arc by specifying a radius and rotation angle. The radius value determines the size of the arc and the rotation angle determines where the arc starts and ends. You need to enter a center point, radius, starting angle and ending angle of the arc.

Diagram C (Fig. 3.5) illustrates some examples of drawing arcs using this method. In this diagram, the center point of the arc is entered at “1” and two angles are specified. For example, to draw the semi-circular arc, 0° is entered as the starting angle and 180° as the ending angle.

2 Points

You can draw a circle or semi-circular arc just by indicating two points. The two points you enter determine the diameter of the circle, and an arc or circle is drawn passing through these points. Diagram D (Fig. 3.5) illustrates some examples of drawing arcs and a circle using this method. Just enter point “1” and “2”, and the arc or circle is drawn.

2 Tangents and a Point

You can draw a circle tangent to two lines and passing through a given point. You need to indicate the two lines and a point within the lines. Diagram A (Fig. 3.6) illustrates an example drawing a circle using this method.

Note:

The point through which the circle passes must be entered inside the two given lines. If it's entered outside the lines, the computer can't draw a circle and an error message to this effect is displayed.

3 Tangents

You can draw a circle tangent to three given lines. Using this method, all you need to do is indicate the three lines. No radius value is required because there can be only one circle tangent to three lines. A circle drawn using this method is shown in Diagram B (Fig. 3.6).

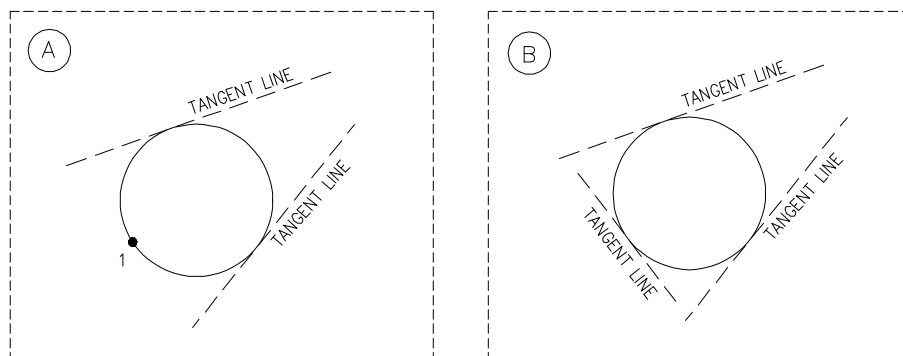


Fig. 3.6: Drawing arcs and circles.

Note:

The number of methods available to draw arcs and circles and the exact procedures used vary from one program to another. Advanced engineering programs include a number of additional methods for drawing arcs and circles. For example, you can draw an arc by specifying its circumference and radius or circumference and rotation angle.

Drawing Ellipses and Elliptical Arcs

Ellipses are much easier to draw with CADD than on a drawing board. On a drawing board, you need to find the right size template or draw a series of arcs individually to draw an ellipse. With CADD, all you need to do is specify the size of the ellipse.

The following are two basic methods for drawing ellipses:

- Length and width
- Axis and rotation angle

Length and Width

An ellipse has two axes: a major axis and a minor axis (Fig. 3.7). The major axis determines the length of the ellipse and the minor axis determines the width.

To draw an ellipse you need to specify length and width, i.e., major axis and minor axis. You can enter the values numerically or by indicating points in the drawing area. When you need to draw an ellipse rotated at an angle, you can indicate the axis rotated at an angle (Fig. 3.7). To draw an elliptical arc, enter the starting point and the end point of the ellipse in addition to indicating the axes.

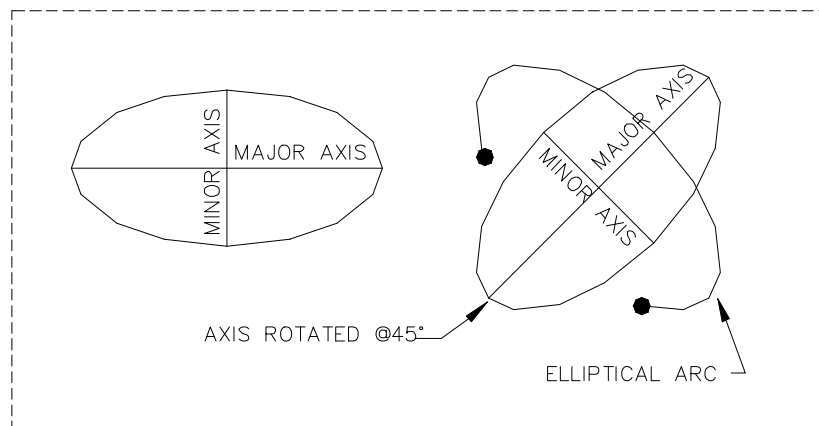


Fig. 3.7: Drawing ellipses by specifying length and width.

Axis and Rotation Angle

When a circle is viewed at an angle, it takes on an elliptical shape. CADD uses the same principle to draw an ellipse. It takes a circle and rotates it into 3D space around one of the axes. As a result, the width of the circle is reduced and an ellipse is drawn (Fig. 3.8).

To draw an ellipse, you need to specify the length of the ellipse (major axis) and the rotation angle. The computer automatically calculates its width and draws an ellipse.

Fig. 3.8 illustrates how a circle creates an ellipse when rotated at different angles. In the first diagram, 0° is entered as the rotation angle; this draws a complete circle. As you enter a higher angle value, the ellipse gets smaller and smaller. Most CADD programs don't allow you to enter a rotation angle greater than 89.4° . This is because the ellipse becomes so narrow that it appears as a line and the computer displays an error message.

Note:

In advanced engineering systems, a number of additional methods for drawing elliptical arcs are available. These methods enable you to draw a variety of elliptical shapes, including a parabola and a hyperbola.

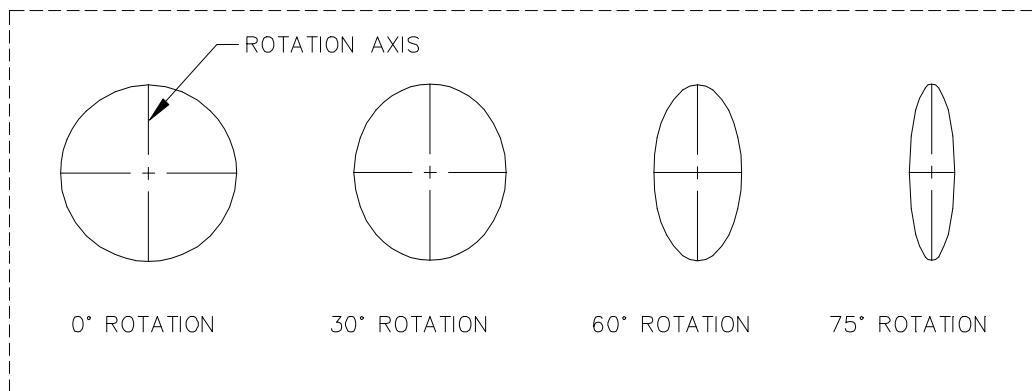


Fig. 3.8: Drawing ellipses with axis and rotation angle method.